

Project Requirement Analysis Phase  
Solution Requirements (Functional & Non-functional)

|               |                                             |
|---------------|---------------------------------------------|
| Date          | 01 July 2026                                |
| Team ID       | SWTID-2026-2152                             |
| Project Name  | Emotion Detection & Learning Support Engine |
| Maximum Marks | 4 Marks                                     |

## Functional Requirements (What the system must do)

### FR-1: Real-Time Multi-Modal Emotion Capture

The system must capture raw video feeds to analyze facial landmarks and micro-expressions for engagement, confusion, frustration, or boredom.

The system must integrate audio feature extraction (tone and pitch) and text-based sentiment analysis from chats or response logs to refine emotion accuracy.

### FR-2: Dynamic Content Adaptation

The learning engine must automatically trigger personalized interventions (e.g., provide hints, adapt instruction pacing, or recommend a short break) when high stress or low engagement is detected.

The system must adjust the difficulty level of curriculum pathways based on the student's current emotional and cognitive state.

### FR-3: Learner and Instructor Reporting Platforms

The system must offer an interactive learner dashboard displaying personal engagement metrics, mental state trends, and academic milestones.

The system must compile real-time, aggregated data for instructor dashboards to flag students who are struggling or disengaged.

## **Non-Functional Requirements (How the system must perform)**

### **NFR-1: Processing Speed & Latency**

The emotion classification pipeline must process input frames and trigger the corresponding learning support response within an end-to-end latency window of < 500ms to guarantee a seamless user experience.

### **NFR-2: Strict Data Privacy and Governance**

To safeguard biometric details and student identities, all processing feeds must utilize secure end-to-end encryption protocols (AES-256 at rest, TLS 1.3 in transit).

The system architecture must fully comply with localized and global student privacy frameworks (e.g., COPPA, GDPR) regarding sensitive metric tracking.

### **NFR-3: Scalability and Concurrent Infrastructure**

The application architecture must scale dynamically to handle up to 10,000 active, simultaneous user streams without degrading response times or frame capture accuracy.

The learning portal must offer an overall service availability and uptime rate of 99.9% during active institution hours.

The Emotion Detection & Learning Support Engine utilizes real-time text-based sentiment analysis to accurately identify a student's emotional state, such as frustration or confusion, directly from their chats and quiz reflections. Based on these detected text inputs, the platform automatically triggers personalized learning interventions, adjusts curriculum difficulty, and provides tailored instructional pacing to match the user's needs. Simultaneously, the system delivers actionable analytics through interactive dashboards, offering students personalized progress tracking and giving instructors real-time insights to quickly support struggling learners. This entire text-driven ecosystem is built on an enterprise-grade infrastructure that guarantees ultra-low processing latency (< 500ms) and secures all user data with advanced encryption while scaling smoothly for 10,000 concurrent users.